
When Does Task Success Certify Agency?

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Abstract

Modern RL agents are commonly evaluated by success which answers: did the agent reach the requested target, did the policy follow the instruction and did the system respond correctly to the cue? In this work, we argue that success alone does not define agency. A target may be reached because the environment passively produces it, many goals may collapse to the same behavior, or cue-conditioned responses may arise from leakage rather than perception. We formalize exactly when success on a task family forces information to pass through the agent-environment interface. Our key object of study is a behaviorally distinguishable task label: a goal label is meaningful only when it can be decoded from the achieved trajectory, and a cue label is meaningful only when it can be decoded from the agent’s response. Under some no-side-channel assumptions, we prove Fano-inequality-style lower bounds showing that achieving M distinguishable goals with high probability requires action-to-observation information flow, or empowerment, of order $\log M$, up to some decoding error. On the flip side, responding to M distinguishable cues with high probability requires observation-to-action information flow, or plasticity, of order $\log M$. Putting these lower bounds together with conditional-entropy and reachable-support upper bounds gives a necessary condition for goal-conditioned behavior: if distinguishable task demand exceeds the interface budget, high-probability success is impossible unless some assumption fails. We also discuss the limitations of viewing RL problems through the lens of empowerment and plasticity, showing that the derived bounds are not always useful through specific MDP constructions.

1 Introduction

Goal conditioned RL agents are often evaluated by success rate which tracks whether the agent reached some requested state, produced some requested configuration, or if it responded correctly to some observed cue. However, success is not the same as agency. For example, target may be reached simply because the environment evolves there regardless of the action sequence. A cue may be answered correctly simply because it leaks through a hidden variable. A benchmark can contain many labels that collapse to the same achieved behavior. This motivates a question: when does success on a task family force genuine information flow through the agent-environment interface?

In this work, we argue that the relevant unit is a family of behaviorally distinguishable alternatives rather than just a single goal. If some achieved agent trajectory reveals which goal was requested, then goal information must have entered future observations only through the action side of the interface. If the future action sequence reveals which cue was observed, then cue information must have entered the policy only through the observation side of the interface. We make this intuition precise using generalized directed information: empowerment for action-to-observation influence and plasticity for observation-to-action influence, building on the framework introduced by Abel et al (1) - our work is largely an expansion of their paper.

Empowerment was introduced as an intrinsic measure of control: an agent is empowered when its actions can create many distinguishable future observations (2). Subsequent work connected

empowerment to directed information in perception-action loops (3), building on feedback and bidirectional information theory (5; 4). Abel et al. recently proposed plasticity as the mirror image of empowerment: instead of measuring how actions affect future observations, plasticity measures how observations affect future actions (1). Their generalized directed-information formalism makes this symmetry precise and proves that the two directions share a finite interface budget.

Our work studies how that tradeoff constrains success-based evaluation and should influence the design of such metrics. A fixed target outcome is too weak to certify agency: if the environment outputs the target trajectory regardless of the action sequence, the agent succeeds with probability one but has no control over the outcome. A raw goal label is also too weak: one can increase $H(G)$ by relabeling goals without changing the agent-environment interaction. The information that matters is the part of the task label that is behaviorally distinguishable. We formalize this by replacing a raw goal G with a coarsened label $U = u(G)$ that can be decoded from the future observation block. On the other hand, environmental cues are represented by a distinguishable cue label $V = v(C)$ that can be decoded from the agent’s future action block.

The paper has three main contributions. First, we prove lower bounds: if U is decodable from future observations with error δ , then the action-to-observation directed information must be at least $H(U)$ up to small correction terms; if V is decodable from future actions with error ε , then the observation-to-action directed information satisfies the mirror lower bound. Second, we refine the upper side of the tradeoff by replacing coarse alphabet-size bounds with conditional-entropy budgets depending on the realized interaction distribution. Third, in finite-horizon MDPs, we find that reachable continuation set sizes control these budgets, while also showing why such bounds can be very impractical in certain MDP instances.

2 Related Works

The empowerment literature looks at agent control as information flow from actions to future states(2). This relates to the action sequence not being entirely independent, as later actions depends on earlier observations. Directed information was introduced to describe this causal dependence. (5; 4). Capdepuy’s work made the notion of empowerment and its connection to causal information flow explicit.(3). Abel et al. extend this by defining plasticity as the opposite direction of generalized directed information and proving a conservation theorem for their sum(1).

Our contribution asks what conservation law says about goals. Goal-directedness is not itself an information-theoretic primitive, so needs to be studied finely. Goals come in a variety of formats, all carrying different amounts of behaviorally relevant information. The distinguishable-label formulation isolates the part that matters for the conservation. Namely, this is the component of the goal that can be recoverable from what ended up happening.

Our paper also relates to classical information-theoretic lower bounds. Fano’s inequality connects decoding goals into a mutual-information requirement (7; 8). Here the decoder is an external observer of the agent-environment trajectory. The lower bound therefore has a different interpretation: if the trajectory must reveal which task label was achieved, then the corresponding direction of the interaction must carry enough information.

Finally, the MDP section relates this work with the reinforcement-learning paradigm. Enforcing stricter conditions like Linear MDP and Bellman-completeness assumptions can make analyzing MDPs using dynamic programming problems tractable, but they are not incompatible with high trajectory agency. The derived bounds, therefore, point to a different class of structural restrictions on MDPs, connected to reachable support and the number of distinguishable continuations after a realized history.

3 Preliminaries

Following (1), we work in the finite-horizon agent-environment framework. Time is discrete with horizon n . The agent and environment exchange symbols from finite alphabets, $\mathcal{A}$ and $\mathcal{O}$, and induce random sequences

$$A_{1:n} = (A_1, \dots, A_n), \quad O_{1:n} = (O_1, \dots, O_n).$$

An agent λ maps histories that end in observations to distributions over actions, while an environment e maps histories that end in actions to distributions over observations. Together, a fixed pair (λ, e) induces the joint law of $A_{1:n}$ and $O_{1:n}$.

Fix two windows $[a : b]$ and $[c : d]$ with

$$1 \leq a \leq b \leq n, \quad 1 \leq c \leq d \leq n.$$

We work in the regime $a \leq c \leq d$, so that the right window begins does not begin earlier than the left one. We allow overlap between the windows.

For arbitrary discrete sequences $X_{1:n}$ and $Y_{1:n}$, (1) defines the generalized directed information $X_{a:b}$ to $Y_{c:d}$ as

$$I(X_{a:b} \rightarrow Y_{c:d}) := \sum_{i=\max(a,c)}^d I(X_{a:\min(b,i)}; Y_i \mid X_{1:a-1}, Y_{1:i-1}).$$

When $a = c = 1$ and $b = d = n$, this reduces to usual definition of directed information, i.e.

$$I(X_{1:n} \rightarrow Y_{1:n}) = \sum_{i=1}^n I(X_{1:i}; Y_i \mid Y_{1:i-1}).$$

We also use the following notation

$$I(X_{a:b} \rightarrow Y_{c:d}) := \sum_{i=\max(a+1,c)}^d I(X_{a:\min(b,i-1)}; Y_i \mid X_{1:a-1}, Y_{1:i-1}),$$

to account for the case where symbols in the right-hand sequence happen after the corresponding symbols in the left-hand sequence.

Empowerment measures action-to-observation influence, while plasticity measures the mirror direction, observation-to-action influence. Following (1), define the windowed empowerment and plasticity of a fixed agent-environment pair (λ, e) by

$$E_{a:b \rightarrow c:d}(\lambda, e) := I(A_{a:b} \rightarrow O_{c:d}), \quad P_{a:b \rightarrow c:d}(\lambda, e) := I(O_{a:b} \rightarrow A_{c:d}).$$

Because $a \leq c$, these generalized directed-information quantities simplify to

$$\begin{aligned} E_{a:b \rightarrow c:d}(\lambda, e) &= \sum_{i=c}^d I(A_{a:\min(b,i)}; O_i \mid A_{1:a-1}, O_{1:i-1}), \\ P_{a:b \rightarrow c:d}(\lambda, e) &= \sum_{i=c}^d I(O_{a:\min(b,i)}; A_i \mid O_{1:a-1}, A_{1:i-1}). \end{aligned}$$

It is helpful to introduce the following shorthands

$$H_{AO}^- := (A_{1:a-1}, O_{1:c-1}), \quad H_{OA}^- := (O_{1:a-1}, A_{1:c-1}).$$

Lastly, the generalized directed-information conservation law of (1) implies the following results, where the first is the empowerment-oriented conservation law and the second is the plasticity-oriented conservation law

$$\begin{aligned} I(A_{a:b}; O_{c:d} \mid H_{AO}^-) &= I(A_{a:b} \rightarrow O_{c:d}) + I(O_{c:d} \rightarrow A_{a:b}), \\ I(O_{a:b}; A_{c:d} \mid H_{OA}^-) &= I(O_{a:b} \rightarrow A_{c:d}) + I(A_{c:d} \rightarrow O_{a:b}). \end{aligned}$$

3.1 Goal and cue families

Let $\mathcal{G}$ be a finite goal set, and let

$$G \sim \mu \in \Delta(\mathcal{G})$$

be sampled initially. Say each goal $g \in \mathcal{G}$ specifies a certain success set $T_g \subseteq \mathcal{O}^{d-c+1}$. Write $\tau := O_{c:d}$. A natural definition of success could be

$$\Pr(\tau \in T_G) \geq 1 - \delta.$$

But because different goals may have overlapping success sets, we introduce a finite distinguishable goal label $U = u(G) \in \mathcal{U}$, together with a decoder $\hat{U} : \mathcal{O}^{d-c+1} \rightarrow \mathcal{U}$, which satisfies $\Pr(\hat{U}(O_{c:d}) \neq U) \leq \delta$.

Symmetrically, let $\mathcal{C}$ be a finite cue set, let $C \sim \nu \in \Delta(\mathcal{C})$, and write $\alpha := A_{c:d}$. A distinguishable cue label is a finite random variable $V = v(C) \in \mathcal{V}$, for which there is a decoder $\hat{V} : \mathcal{A}^{d-c+1} \rightarrow \mathcal{V}$, such that $\Pr(\hat{V}(A_{c:d}) \neq V) \leq \varepsilon$.

3.2 No-side-channel assumptions

The next two conditions rule out a shortcut: the task should not be given away directly. If future observations or actions reveal the task, that information has to come through the interaction history.

Assumption 3.1 (Goal-to-observation no-side-channel). *For every $i \in \{c, \dots, d\}$,*

$$I(U; O_i \mid A_{a:\min(b,i)}, A_{1:a-1}, O_{1:i-1}) = 0.$$

Equivalently,

$$U \longrightarrow (A_{a:\min(b,i)}, A_{1:a-1}, O_{1:i-1}) \longrightarrow O_i$$

is a Markov chain.

Assumption 3.2 (Cue-to-action no-side-channel). *For every $i \in \{c, \dots, d\}$,*

$$I(V; A_i \mid O_{a:\min(b,i)}, O_{1:a-1}, A_{1:i-1}) = 0.$$

Equivalently,

$$V \longrightarrow (O_{a:\min(b,i)}, O_{1:a-1}, A_{1:i-1}) \longrightarrow A_i$$

is a Markov chain.

Finally, let

$$h_2(p) := -p \log p - (1-p) \log(1-p)$$

mean the binary entropy function. Unless otherwise stated, all logarithms are base 2.

4 Distinguishable Tasks Force Information Flow

This section gives lower bounds on empowerment and plasticity in terms of information about goals and cues. If future observations allow us to recover a goal variable U , then those observations must have some information about U . Under the no-side-channel assumption, that information must be carried by the agent's actions, giving a lower bound on empowerment. The cue case is the same argument with the roles reversed.

4.1 Structural information lower bounds

The following two results compare the task information in the future block with the directed information across the same window.

Theorem 4.1. *Under Assumption 3.1,*

$$E_{a:b \rightarrow c:d}(\lambda, e) \geq I(U; O_{c:d} \mid H_{AO}^-).$$

Theorem 4.2. *Under Assumption 3.2,*

$$P_{a:b \rightarrow c:d}(\lambda, e) \geq I(V; A_{c:d} \mid H_{OA}^-).$$

4.2 Conditional bound

This lemma connects decoding to entropy. If Y is enough to recover X with small error, then X cannot have much uncertainty after Y is known. The variable Z is useful because we will later condition on the earlier history.

Lemma 4.3. *Let X be a finite-valued random variable taking values in $\mathcal{X}$ with $|\mathcal{X}| > 1$, let Y and Z be arbitrary random variables, let $\eta \in [0, \frac{1}{2}]$, and let $\hat{X} = \phi(Y)$ be any decoder with*

$$\Pr(\hat{X} \neq X) \leq \eta.$$

Then

$$H(X \mid Y, Z) \leq h_2(\eta) + \eta \log(|\mathcal{X}| - 1).$$

4.3 Decoding corollaries

These corollaries consider the decoder. If U can be recovered from $O_{c:d}$, then $O_{c:d}$ must contain almost all of the remaining uncertainty in U . Combined with the structural information lower bound above, it forces empowerment to be large. A similar argument gives the plasticity bound.

Corollary 4.4. *Assume Assumption 3.1, suppose U takes values in $\mathcal{U}$ with $|\mathcal{U}| > 1$, suppose $\delta \in [0, \frac{1}{2}]$, and suppose U has a decoder $\hat{U}(O_{c:d})$ with*

$$\Pr(\hat{U}(O_{c:d}) \neq U) \leq \delta.$$

Then

$$E_{a:b \rightarrow c:d}(\lambda, e) \geq H(U \mid H_{AO}^-) - h_2(\delta) - \delta \log(|\mathcal{U}| - 1).$$

Corollary 4.5. *Assume Assumption 3.2, suppose V takes values in $\mathcal{V}$ with $|\mathcal{V}| > 1$, suppose $\varepsilon \in [0, \frac{1}{2}]$, and suppose V has a decoder $\hat{V}(A_{c:d})$ with*

$$\Pr(\hat{V}(A_{c:d}) \neq V) \leq \varepsilon.$$

Then

$$P_{a:b \rightarrow c:d}(\lambda, e) \geq H(V \mid H_{OA}^-) - h_2(\varepsilon) - \varepsilon \log(|\mathcal{V}| - 1).$$

4.4 Fully distinguishable special case

The previous results were about the most general setting, considering only the part of G that can be recovered from the future observations. We now consider the simplest case where goals themselves are distinguishable.

Proposition 4.6. *Assume the success sets $\{T_g\}_{g \in \mathcal{G}}$ are pairwise disjoint and $\Pr(O_{c:d} \in T_G) \geq 1 - \delta$. Then G admits a decoder from $O_{c:d}$ with error at most δ .*

Corollary 4.7. *Assume:*

1. *the success sets $\{T_g\}_{g \in \mathcal{G}}$ are pairwise disjoint and $\Pr(O_{c:d} \in T_G) \geq 1 - \delta$;*
2. *C has a decoder $\hat{C}(A_{c:d})$ with $\Pr(\hat{C}(A_{c:d}) \neq C) \leq \varepsilon$;*
3. *Assumption 3.1 holds with $U = G$;*
4. *Assumption 3.2 holds with $V = C$;*
5. *$|\mathcal{G}| > 1$, $|\mathcal{C}| > 1$, $\delta, \varepsilon \in [0, \frac{1}{2}]$;*
6. *G is independent of H_{AO}^- , and C is independent of H_{OA}^- .*

Then

$$E_{a:b \rightarrow c:d}(\lambda, e) \geq H(G) - h_2(\delta) - \delta \log(|\mathcal{G}| - 1),$$

and

$$P_{a:b \rightarrow c:d}(\lambda, e) \geq H(C) - h_2(\varepsilon) - \varepsilon \log(|\mathcal{C}| - 1).$$

The full proofs of Theorems 4.1-4.2 and Corollaries 4.4-4.7 are in Appendix A.

5 Interface Budgets and Impossibility Tests

We give sharper upper bounds on $E_{a:b \rightarrow c:d}(\lambda, e)$ and $P_{a:b \rightarrow c:d}(\lambda, e)$. We then combine those upper bounds with the lower bounds from Section 4. If the task requires more information than these bounds allow, then it cannot be achieved under the stated assumptions.

5.1 Conditional information budgets

Define

$$B_{a:b,c:d}^{OA}(\lambda, e) := I(O_{a:b}; A_{c:d} \mid H_{OA}^-),$$

and

$$B_{a:b,c:d}^{AO}(\lambda, e) := I(A_{a:b}; O_{c:d} \mid H_{AO}^-).$$

These are the total conditional informations available in each direction, with the first bounding observation-to-action information and the second bounding action-to-observation information.

Proposition 5.1. *For every agent-environment pair (λ, e) ,*

$$P_{a:b \rightarrow c:d}(\lambda, e) \leq B_{a:b,c:d}^{OA}(\lambda, e) \leq m_{\text{ent}}^{OA}(\lambda, e),$$

where

$$m_{\text{ent}}^{OA}(\lambda, e) := \min \left\{ H(O_{a:b} \mid H_{OA}^-), H(A_{c:d} \mid H_{OA}^-) \right\}.$$

Similarly,

$$E_{a:b \rightarrow c:d}(\lambda, e) \leq B_{a:b,c:d}^{AO}(\lambda, e) \leq m_{\text{ent}}^{AO}(\lambda, e),$$

where

$$m_{\text{ent}}^{AO}(\lambda, e) := \min \left\{ H(A_{a:b} \mid H_{AO}^-), H(O_{c:d} \mid H_{AO}^-) \right\}.$$

Now we show that conditional entropy bounds imply the original alphabet-size upper bounds from (1).

Corollary 5.2. *For every (λ, e) ,*

$$m_{\text{ent}}^{OA}(\lambda, e) \leq \min \left\{ (b - a + 1) \log |\mathcal{O}|, (d - c + 1) \log |\mathcal{A}| \right\},$$

$$m_{\text{ent}}^{AO}(\lambda, e) \leq \min \left\{ (b - a + 1) \log |\mathcal{A}|, (d - c + 1) \log |\mathcal{O}| \right\}.$$

Consequently,

$$P_{a:b \rightarrow c:d}(\lambda, e) \leq \min \left\{ (b - a + 1) \log |\mathcal{O}|, (d - c + 1) \log |\mathcal{A}| \right\},$$

$$E_{a:b \rightarrow c:d}(\lambda, e) \leq \min \left\{ (b - a + 1) \log |\mathcal{A}|, (d - c + 1) \log |\mathcal{O}| \right\}.$$

5.2 Impossibility tests

Combining the lower bounds from Section 4 with the upper bounds considered above will give us the necessary conditions for a task family to be achievable under the no-side-channel assumptions. A task may require more distinguishable information than the chosen interface can carry, and when that happens, it is impossible under the stated assumptions.

Corollary 5.3. *Assume the hypotheses of Corollary 4.4 for the distinguishable goal variable U . Then*

$$H(U \mid H_{AO}^-) - h_2(\delta) - \delta \log(|\mathcal{U}| - 1) \leq m_{\text{ent}}^{AO}(\lambda, e).$$

Assume the hypotheses of Corollary 4.5 for the distinguishable cue variable V . Then

$$H(V \mid H_{OA}^-) - h_2(\varepsilon) - \varepsilon \log(|\mathcal{V}| - 1) \leq m_{\text{ent}}^{OA}(\lambda, e).$$

Define

$$D_{\text{goal}} := H(U \mid H_{AO}^-) - h_2(\delta) - \delta \log(|\mathcal{U}| - 1), \quad B_{\text{goal}} := m_{\text{ent}}^{AO}(\lambda, e).$$

Then goal achievement under the no-side-channel assumption requires $D_{\text{goal}} \leq B_{\text{goal}}$. Similarly, define

$$D_{\text{cue}} := H(V \mid H_{OA}^-) - h_2(\varepsilon) - \varepsilon \log(|\mathcal{V}| - 1), \quad B_{\text{cue}} := m_{\text{ent}}^{OA}(\lambda, e).$$

Cue-conditioned behavior under the no-side-channel assumption requires $D_{\text{cue}} \leq B_{\text{cue}}$.

Figure 1 illustrates the condition $D_{\text{goal}} \leq B_{\text{goal}}$ empirically in a small grid world we simulated.

We can think of D_{goal} and D_{cue} as information demands forced by distinguishability requirement. The quantities B_{goal} and B_{cue} are the corresponding information budgets for the chosen windows. If a demand exceeds its budget, then the assumptions cannot all be true at the claimed error level.

The full proofs of Proposition 5.1, Corollary 5.2, and Corollary 5.3 are in Appendix B.

In the following example, we illustrate the failure of the condition $D_{\text{goal}} \leq B_{\text{goal}}$ theoretically.

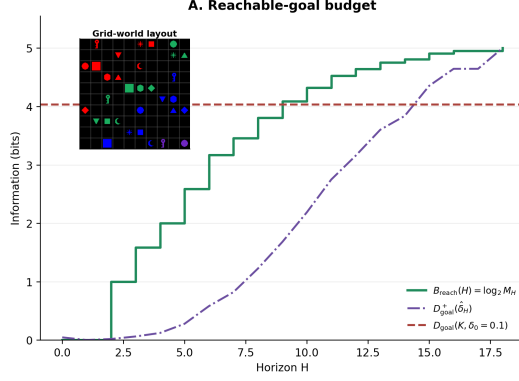


Figure 1: Grid-world illustration. (A) As the horizon grows, more target docks become reachable. The reachable-goal budget is

$$B_{\text{reach}}(H) = \log_2 M_H,$$

where M_H is the number of docks reachable within horizon H . The purple curve shows

$$D_{\text{goal}}^+(\hat{\delta}_H) = \max\{0, \log_2 K - h_2(\hat{\delta}_H) - \hat{\delta}_H \log_2(K-1)\}.$$

The red line is the demand at target error tolerance $\delta_0 = 0.1$. All information quantities are in bits.

5.3 Example: four goals

We will sanity check the impossibility condition with a very simple setting. Consider a task with four equally likely goal values,

$$U \in \{1, 2, 3, 4\}, \quad H(U) = 2.$$

Assume U is independent of H_{AO}^- . Assume the future observation block takes only two values, $\{0, 1\}$, i.e. $O_{c:d}$ contains at most one bit. Take the error threshold to be $\delta = 0.1$. Since any decoder from $O_{c:d}$ has only two possible inputs, it can identify at most two of the four goal values. Since U is uniform, the error probability is at least $1/2$, so error 0.1 is impossible.

Our impossibility test supports this intuition. The goal demand is

$$D_{\text{goal}} = H(U | H_{AO}^-) - h_2(0.1) - 0.1 \log 3 = 2 - h_2(0.1) - 0.1 \log 3 \approx 1.37.$$

On the other hand,

$$B_{\text{goal}} = m_{\text{ent}}^{AO}(\lambda, e) \leq H(O_{c:d} | H_{AO}^-) \leq 1.$$

Therefore $D_{\text{goal}} > B_{\text{goal}}$.

So, no agent-environment pair under the no-side-channel assumption can make four uniformly likely goals distinguishable from a binary future observation with error at most 0.1.

6 Reachable-Continuation Budgets in MDPs

What if we restrict our setting from the most general one to a finite-horizon, fully observed MDP? We can show that the plasticity-empowerment framework becomes sensitive to the size of the reachable continuation tree.

6.1 Finite-horizon MDP setting

Let $\mathcal{S}$ be a finite set of states and $\mathcal{A}$ a finite set of actions. We consider a finite-horizon, fully observed MDP with horizon n , initial-state distribution ρ , transition kernels

$$P_t(\cdot | s, a) \in \Delta(\mathcal{S}), \quad t = 1, \dots, n-1,$$

and a possibly nonstationary policy

$$\pi_t(\cdot | s) \in \Delta(\mathcal{A}), \quad t = 1, \dots, n.$$

Note that

$$S_1 \sim \rho, \quad A_t | S_t \sim \pi_t(\cdot | S_t), \quad S_{t+1} | (S_t, A_t) \sim P_t(\cdot | S_t, A_t).$$

We identify observations with states, $O_t = S_t$. We also restrict to strictly future windows,

$$1 \leq a \leq b < c \leq d \leq n,$$

i.e. $A_{a:b}$ happens before $S_{c:d}$ and $S_{a:b}$ happens before $A_{c:d}$. Define the following useful shortcuts

$$\tilde{H}^- := (A_{1:a-1}, S_{1:c-1}), \quad H^- := (S_{1:a-1}, A_{1:c-1}).$$

Define the lower-bound quantities

$$L_U := H(U | \tilde{H}^-) - h_2(\delta) - \delta \log(|\mathcal{U}| - 1), \quad L_V := H(V | H^-) - h_2(\varepsilon) - \varepsilon \log(|\mathcal{V}| - 1).$$

Recall that under hypotheses of Corollaries 4.4 and 4.5,

$$L_U \leq E_{a:b \rightarrow c:d}(\lambda, e), \quad L_V \leq P_{a:b \rightarrow c:d}(\lambda, e).$$

6.2 Reachable continuation sets

For a realized history $\tilde{H}^- = \tilde{h}^-$, define the reachable future-state block

$$\mathcal{R}_S(\tilde{h}^-) := \left\{ s_{c:d} \in \mathcal{S}^{d-c+1} : \Pr(S_{c:d} = s_{c:d} | \tilde{H}^- = \tilde{h}^-) > 0 \right\}.$$

For a realized history $H^- = h^-$, define the reachable future-action block

$$\mathcal{R}_A(h^-) := \left\{ a_{c:d} \in \mathcal{A}^{d-c+1} : \Pr(A_{c:d} = a_{c:d} | H^- = h^-) > 0 \right\}.$$

The following lemma is a general result that says that the uncertainty left after conditioning has an upper bound that is a function of the number of outcomes that remain possible. In the MDP, the remaining possible outcomes are the reachable future state and action blocks, so applying the lemma to our setting results in Proposition 6.2.

Lemma 6.1. *Let X and Z be discrete random variables. Suppose for all z in the support of Z ,*

$$|\text{supp}(X | Z = z)| \leq M(z)$$

for some deterministic $M(z) \geq 1$. Then

$$H(X | Z) \leq \mathbb{E}[\log M(Z)].$$

In particular, if $M(z) \leq M$ uniformly for all z , then

$$H(X | Z) \leq \log M.$$

Proposition 6.2 (Reachable-continuation bound). *For finite-horizon fully observed MDPs, we have*

$$E_{a:b \rightarrow c:d}(\lambda, e) \leq \mathbb{E}[\log |\mathcal{R}_S(\tilde{H}^-)|], \quad P_{a:b \rightarrow c:d}(\lambda, e) \leq \mathbb{E}[\log |\mathcal{R}_A(H^-)|].$$

If there are uniform bounds N_S, N_A such that $|\mathcal{R}_S(\tilde{h}^-)| \leq N_S$ and $|\mathcal{R}_A(h^-)| \leq N_A$ for all reachable $\tilde{h}^-, h^-$, then

$$E_{a:b \rightarrow c:d}(\lambda, e) \leq \log N_S, \quad P_{a:b \rightarrow c:d}(\lambda, e) \leq \log N_A.$$

Under the hypotheses of the lower bounds from Section 4, these results further imply

$$L_U \leq \mathbb{E}[\log |\mathcal{R}_S(\tilde{H}^-)|], \quad L_V \leq \mathbb{E}[\log |\mathcal{R}_A(H^-)|],$$

and in particular

$$L_U \leq \log N_S, \quad L_V \leq \log N_A.$$

In the simple interface setting of (1), empowerment and plasticity are bounded by the size of the full alphabet, which provides stronger upper bounds and recovers the original ones.

Proposition 6.3. *For finite-horizon fully observed MDPs,*

$$E_{a:b \rightarrow c:d}(\lambda, e) \leq \min \left\{ (b - a + 1) \log |\mathcal{A}|, \mathbb{E}[\log |\mathcal{R}_S(\tilde{H}^-)|] \right\},$$

$$P_{a:b \rightarrow c:d}(\lambda, e) \leq \min \left\{ (b - a + 1) \log |\mathcal{S}|, \mathbb{E}[\log |\mathcal{R}_A(H^-)|] \right\}.$$

Moreover,

$$\mathbb{E}[\log |\mathcal{R}_S(\tilde{H}^-)|] \leq (d - c + 1) \log |\mathcal{S}|,$$

$$\mathbb{E}[\log |\mathcal{R}_A(H^-)|] \leq (d - c + 1) \log |\mathcal{A}|.$$

Therefore,

$$E_{a:b \rightarrow c:d}(\lambda, e) \leq \min \left\{ (b - a + 1) \log |\mathcal{A}|, (d - c + 1) \log |\mathcal{S}| \right\},$$

$$P_{a:b \rightarrow c:d}(\lambda, e) \leq \min \left\{ (b - a + 1) \log |\mathcal{S}|, (d - c + 1) \log |\mathcal{A}| \right\}.$$

6.3 When are the bounds loose?

We will provide a counterexample that show that our framework is not always useful and does not always lead to sharper bounds.

Example 6.4 (Action-independent uniform-noise MDP). *Fix $|\mathcal{S}| \geq 2$, $|\mathcal{A}| \geq 2$, and horizon $n \geq 2$. Let*

$$\rho(s) = \frac{1}{|\mathcal{S}|}, \quad P_t(s' | s, a) = \frac{1}{|\mathcal{S}|},$$

for all $s, s' \in \mathcal{S}$, $a \in \mathcal{A}$, t . Let the policy be uniform and independent of the state, $\pi_t(a | s) = 1/|\mathcal{A}|$, for all $s \in \mathcal{S}$, $a \in \mathcal{A}$, t . As before, identify observations with states, $O_t = S_t$.

For all choices of strictly future windows, i.e.

$$1 \leq a \leq b < c \leq d \leq n,$$

we have

$$E_{a:b \rightarrow c:d}(\lambda, e) = 0, \quad P_{a:b \rightarrow c:d}(\lambda, e) = 0,$$

while

$$\mathbb{E}[\log |\mathcal{R}_S(\tilde{H}^-)|] = (d - c + 1) \log |\mathcal{S}|, \quad \mathbb{E}[\log |\mathcal{R}_A(H^-)|] = (d - c + 1) \log |\mathcal{A}|.$$

This example separates reachability from influence: every state and action are reachable, but actions do not affect future states and states do not affect future actions, so empowerment and plasticity are both zero. Moreover, the expectation of the log-size of the reachable set is then equal to the coarse term from (1).

The full proofs of Lemma 6.1, Proposition 6.2, Proposition 6.3, and Example 6.4 are in Appendix C.

7 Discussion and Limitations

The results we've seen provide necessary conditions, but these are not sufficient. It is possible for a system to have high empowerment without pursuing a useful goal and have high plasticity without reacting appropriately, showing the limitations of the framework. Furthermore, in many environments, computing an exact directed-information estimate is infeasible and provides more of a conceptual basis for analysis. Namely, we can identify distinguishable task information to upper-bound information the environment can transmit. Utilizing this, we can verify whether success in an environment could be due to agent control and adaptation, rather than a guaranteed outcome.

A Proofs for Section 4

Proof of Theorem 4.1. For all $i \in \{c, \dots, d\}$, define

$$Z_i := (A_{1:a-1}, O_{1:i-1}), \quad X_i := A_{a:\min(b,i)}, \quad Y_i := O_i.$$

The i -th term in the expansion of $E_{a:b \rightarrow c:d}(\lambda, e)$ is

$$I(X_i; Y_i \mid Z_i).$$

By Assumption 3.1,

$$I(U; Y_i \mid X_i, Z_i) = 0.$$

Apply the chain rule:

$$I(U, X_i; Y_i \mid Z_i) = I(X_i; Y_i \mid Z_i) + I(U; Y_i \mid X_i, Z_i),$$

$$I(U, X_i; Y_i \mid Z_i) = I(U; Y_i \mid Z_i) + I(X_i; Y_i \mid U, Z_i).$$

Using $I(U; Y_i \mid X_i, Z_i) = 0$, the first identity becomes

$$I(U, X_i; Y_i \mid Z_i) = I(X_i; Y_i \mid Z_i).$$

Using nonnegativity in the second identity,

$$I(U, X_i; Y_i \mid Z_i) \geq I(U; Y_i \mid Z_i).$$

Hence

$$I(X_i; Y_i \mid Z_i) \geq I(U; Y_i \mid Z_i),$$

ie.,

$$I(A_{a:\min(b,i)}; O_i \mid A_{1:a-1}, O_{1:i-1}) \geq I(U; O_i \mid A_{1:a-1}, O_{1:i-1}).$$

Summing over $i = c, \dots, d$,

$$E_{a:b \rightarrow c:d}(\lambda, e) = \sum_{i=c}^d I(A_{a:\min(b,i)}; O_i \mid A_{1:a-1}, O_{1:i-1}) \geq \sum_{i=c}^d I(U; O_i \mid A_{1:a-1}, O_{1:i-1}).$$

By the chain rule for conditional mutual information,

$$I(U; O_{c:d} \mid A_{1:a-1}, O_{1:c-1}) = \sum_{i=c}^d I(U; O_i \mid A_{1:a-1}, O_{1:i-1}).$$

So

$$E_{a:b \rightarrow c:d}(\lambda, e) \geq I(U; O_{c:d} \mid H_{AO}^-).$$

□

Proof of Theorem 4.2. For all $i \in \{c, \dots, d\}$, define

$$Z_i := (O_{1:a-1}, A_{1:i-1}), \quad X_i := O_{a:\min(b,i)}, \quad Y_i := A_i.$$

The i -th term in the expansion of $P_{a:b \rightarrow c:d}(\lambda, e)$ is

$$I(X_i; Y_i \mid Z_i).$$

By Assumption 3.2,

$$I(V; Y_i \mid X_i, Z_i) = 0.$$

Apply the chain rule:

$$I(V, X_i; Y_i \mid Z_i) = I(X_i; Y_i \mid Z_i) + I(V; Y_i \mid X_i, Z_i),$$

$$I(V, X_i; Y_i \mid Z_i) = I(V; Y_i \mid Z_i) + I(X_i; Y_i \mid V, Z_i).$$

Using $I(V; Y_i \mid X_i, Z_i) = 0$ and nonnegativity gives

$$I(X_i; Y_i \mid Z_i) \geq I(V; Y_i \mid Z_i),$$

which means

$$I(O_{a:\min(b,i)}; A_i \mid O_{1:a-1}, A_{1:i-1}) \geq I(V; A_i \mid O_{1:a-1}, A_{1:i-1}).$$

Summing over $i = c, \dots, d$,

$$P_{a:b \rightarrow c:d}(\lambda, e) = \sum_{i=c}^d I(O_{a:\min(b,i)}; A_i \mid O_{1:a-1}, A_{1:i-1}) \geq \sum_{i=c}^d I(V; A_i \mid O_{1:a-1}, A_{1:i-1}).$$

By the chain rule,

$$I(V; A_{c:d} \mid O_{1:a-1}, A_{1:c-1}) = \sum_{i=c}^d I(V; A_i \mid O_{1:a-1}, A_{1:i-1}).$$

Finally

$$P_{a:b \rightarrow c:d}(\lambda, e) \geq I(V; A_{c:d} \mid H_{OA}^-).$$

□

Proof of Lemma 4.3. Let

$$E := \mathbf{1}\{\hat{X} \neq X\}$$

be the indicator variable of the error of decoding. Since $\hat{X}$ is a function of Y , the variable E is determined by (X, Y) , so also by (X, Y, Z) , so

$$H(E \mid X, Y, Z) = 0.$$

Now,

$$H(X \mid Y, Z) = H(E, X \mid Y, Z) - H(E \mid X, Y, Z) = H(E, X \mid Y, Z).$$

Via the chain rule,

$$H(E, X \mid Y, Z) = H(E \mid Y, Z) + H(X \mid E, Y, Z),$$

so

$$H(X \mid Y, Z) = H(E \mid Y, Z) + H(X \mid E, Y, Z).$$

For the first term, conditioning cannot increase entropy, so

$$H(E \mid Y, Z) \leq H(E).$$

Let

$$p_e := \Pr(E = 1).$$

Then

$$H(E) = h_2(p_e).$$

Since $p_e \leq \eta$ and $\eta \in [0, \frac{1}{2}]$, and since h_2 is increasing on $[0, \frac{1}{2}]$,

$$H(E) = h_2(p_e) \leq h_2(\eta).$$

Hence

$$H(E \mid Y, Z) \leq h_2(\eta).$$

For the second term,

$$H(X \mid E, Y, Z) = \Pr(E = 0) H(X \mid E = 0, Y, Z) + \Pr(E = 1) H(X \mid E = 1, Y, Z).$$

If $E = 0$, then $X = \hat{X} = \phi(Y)$, so

$$H(X \mid E = 0, Y, Z) = 0.$$

If $E = 1$, then $X \neq \hat{X}$. Since $\hat{X}$ is determined by Y , the value of X under $E = 1$ can take value in at most $|\mathcal{X}| - 1$ options, so

$$H(X \mid E = 1, Y, Z) \leq \log(|\mathcal{X}| - 1).$$

It follows that

$$H(X \mid E, Y, Z) \leq p_e \log(|\mathcal{X}| - 1) \leq \eta \log(|\mathcal{X}| - 1).$$

Combining the two we get

$$H(X \mid Y, Z) \leq h_2(\eta) + \eta \log(|\mathcal{X}| - 1).$$

□

Proof of Corollary 4.4. By Theorem 4.1,

$$E_{a:b \rightarrow c:d}(\lambda, e) \geq I(U; O_{c:d} \mid H_{AO}^-).$$

Expand

$$I(U; O_{c:d} \mid H_{AO}^-) = H(U \mid H_{AO}^-) - H(U \mid O_{c:d}, H_{AO}^-),$$

and then apply Lemma 4.3 with

$$X = U, \quad Y = O_{c:d}, \quad Z = H_{AO}^-, \quad \eta = \delta.$$

Since $\hat{U}$ depends only on $O_{c:d}$, it is a valid decoder in the presence of some side information H_{AO}^- . So

$$H(U \mid O_{c:d}, H_{AO}^-) \leq h_2(\delta) + \delta \log(|\mathcal{U}| - 1).$$

Substitute and get

$$E_{a:b \rightarrow c:d}(\lambda, e) \geq H(U \mid H_{AO}^-) - h_2(\delta) - \delta \log(|\mathcal{U}| - 1). \quad \square$$

Proof of Corollary 4.5. By Theorem 4.2,

$$P_{a:b \rightarrow c:d}(\lambda, e) \geq I(V; A_{c:d} \mid H_{OA}^-).$$

Expand

$$I(V; A_{c:d} \mid H_{OA}^-) = H(V \mid H_{OA}^-) - H(V \mid A_{c:d}, H_{OA}^-),$$

and then apply Lemma 4.3 with

$$X = V, \quad Y = A_{c:d}, \quad Z = H_{OA}^-, \quad \eta = \varepsilon.$$

So

$$H(V \mid A_{c:d}, H_{OA}^-) \leq h_2(\varepsilon) + \varepsilon \log(|\mathcal{V}| - 1).$$

Substitute and get

$$P_{a:b \rightarrow c:d}(\lambda, e) \geq H(V \mid H_{OA}^-) - h_2(\varepsilon) - \varepsilon \log(|\mathcal{V}| - 1). \quad \square$$

Proof of Proposition 4.6. Let

$$S := \{O_{c:d} \in T_G\}$$

be the success event. By assumption

$$\Pr(S) \geq 1 - \delta.$$

On event S , there exists some g such that $O_{c:d} \in T_g$. Since the success sets are disjoint, this g is unique. By definition of S , such g must be G . Therefore on S , the decoder that outputs the unique g with $O_{c:d} \in T_g$ recovers G exactly. Hence the error event is contained in S^c , and

$$\Pr(\hat{G}(O_{c:d}) \neq G) \leq \Pr(S^c) \leq \delta. \quad \square$$

Proof of Corollary 4.7. By Proposition 4.6, the hypotheses means that G has a decoder from $O_{c:d}$ with error at most δ . So Corollary 4.4 applies with $U = G$, giving

$$E_{a:b \rightarrow c:d}(\lambda, e) \geq H(G \mid H_{AO}^-) - h_2(\delta) - \delta \log(|\mathcal{G}| - 1).$$

We assumed independence of G from H_{AO}^- , so

$$H(G \mid H_{AO}^-) = H(G),$$

so

$$E_{a:b \rightarrow c:d}(\lambda, e) \geq H(G) - h_2(\delta) - \delta \log(|\mathcal{G}| - 1).$$

Similarly, Corollary 4.5 is applies with $V = C$, so

$$P_{a:b \rightarrow c:d}(\lambda, e) \geq H(C \mid H_{OA}^-) - h_2(\varepsilon) - \varepsilon \log(|\mathcal{C}| - 1).$$

We assumed independence of C from H_{OA}^- , so

$$H(C \mid H_{OA}^-) = H(C),$$

so

$$P_{a:b \rightarrow c:d}(\lambda, e) \geq H(C) - h_2(\varepsilon) - \varepsilon \log(|\mathcal{C}| - 1). \quad \square$$

B Proofs for Section 5

Proof of Proposition 5.1. By the generalized directed-information identity,

$$I(O_{a:b}; A_{c:d} \mid H_{OA}^-) = I(O_{a:b} \rightarrow A_{c:d}) + I(A_{c:d} \rightarrow O_{a:b}).$$

The first term is

$$I(O_{a:b} \rightarrow A_{c:d}) = P_{a:b \rightarrow c:d}(\lambda, e),$$

and the second term is nonnegative, which means

$$B_{a:b,c:d}^{OA}(\lambda, e) \geq P_{a:b \rightarrow c:d}(\lambda, e).$$

Now use the fact

$$I(X; Y \mid Z) = H(X \mid Z) - H(X \mid Y, Z),$$

with

$$X = O_{a:b}, \quad Y = A_{c:d}, \quad Z = H_{OA}^-,$$

to get

$$B_{a:b,c:d}^{OA}(\lambda, e) = H(O_{a:b} \mid H_{OA}^-) - H(O_{a:b} \mid A_{c:d}, H_{OA}^-) \leq H(O_{a:b} \mid H_{OA}^-).$$

Symmetrically,

$$B_{a:b,c:d}^{OA}(\lambda, e) = H(A_{c:d} \mid H_{OA}^-) - H(A_{c:d} \mid O_{a:b}, H_{OA}^-) \leq H(A_{c:d} \mid H_{OA}^-).$$

Combining,

$$B_{a:b,c:d}^{OA}(\lambda, e) \leq \min \{H(O_{a:b} \mid H_{OA}^-), H(A_{c:d} \mid H_{OA}^-)\} = m_{\text{ent}}^{OA}(\lambda, e).$$

The other action-to-observation direction is similar. □

Proof of Corollary 5.2. Conditioning cannot increase entropy, so

$$H(O_{a:b} \mid H_{OA}^-) \leq H(O_{a:b}).$$

The block $O_{a:b}$ takes values in $\mathcal{O}^{b-a+1}$, so

$$H(O_{a:b}) \leq \log |\mathcal{O}^{b-a+1}| = (b-a+1) \log |\mathcal{O}|.$$

Similarly,

$$H(A_{c:d} \mid H_{OA}^-) \leq H(A_{c:d}) \leq \log |\mathcal{A}^{d-c+1}| = (d-c+1) \log |\mathcal{A}|.$$

Therefore

$$m_{\text{ent}}^{OA}(\lambda, e) \leq \min \{(b-a+1) \log |\mathcal{O}|, (d-c+1) \log |\mathcal{A}|\}.$$

With Proposition 5.1, this gives

$$P_{a:b \rightarrow c:d}(\lambda, e) \leq \min \{(b-a+1) \log |\mathcal{O}|, (d-c+1) \log |\mathcal{A}|\}.$$

The action-to-observation statement is similar with A and O interchanged. □

Proof of Corollary 5.3. By Corollary 4.4,

$$H(U \mid H_{AO}^-) - h_2(\delta) - \delta \log(|\mathcal{U}| - 1) \leq E_{a:b \rightarrow c:d}(\lambda, e).$$

By Proposition 5.1,

$$E_{a:b \rightarrow c:d}(\lambda, e) \leq m_{\text{ent}}^{AO}(\lambda, e).$$

Combining them means

$$H(U \mid H_{AO}^-) - h_2(\delta) - \delta \log(|\mathcal{U}| - 1) \leq m_{\text{ent}}^{AO}(\lambda, e).$$

Similarly, by Corollary 4.5,

$$H(V \mid H_{OA}^-) - h_2(\varepsilon) - \varepsilon \log(|\mathcal{V}| - 1) \leq P_{a:b \rightarrow c:d}(\lambda, e),$$

and by Proposition 5.1,

$$P_{a:b \rightarrow c:d}(\lambda, e) \leq m_{\text{ent}}^{OA}(\lambda, e).$$

Combine to get

$$H(V \mid H_{OA}^-) - h_2(\varepsilon) - \varepsilon \log(|\mathcal{V}| - 1) \leq m_{\text{ent}}^{OA}(\lambda, e). □$$

C Proofs for Section 6

Proof of Lemma 6.1. Pick fixed z . By assumption $X \mid Z = z$ is supported on at most $M(z)$ points, so its entropy is at most the entropy of the uniform distribution on $M(z)$ points:

$$H(X \mid Z = z) \leq \log M(z).$$

Taking expectation over Z

$$H(X \mid Z) = \mathbb{E}[H(X \mid Z = z)] \leq \mathbb{E}[\log M(Z)].$$

If $M(z) \leq M$ uniformly, then $\log M(Z) \leq \log M$, hence

$$H(X \mid Z) \leq \log M. \quad \square$$

Proof of Proposition 6.2. From Proposition 5.1,

$$E_{a:b \rightarrow c:d}(\lambda, e) \leq m_{\text{ent}}^{AO}(\lambda, e) = \min \left\{ H(A_{a:b} \mid \tilde{H}^-), H(S_{c:d} \mid \tilde{H}^-) \right\},$$

so

$$E_{a:b \rightarrow c:d}(\lambda, e) \leq H(S_{c:d} \mid \tilde{H}^-).$$

By definition, conditional on history $\tilde{H}^- = \tilde{h}^-$, the block $S_{c:d}$ is supported on $\mathcal{R}_S(\tilde{h}^-)$. Therefore Lemma 6.1 gives

$$H(S_{c:d} \mid \tilde{H}^-) \leq \mathbb{E}[\log |\mathcal{R}_S(\tilde{H}^-)|].$$

Combining them:

$$E_{a:b \rightarrow c:d}(\lambda, e) \leq \mathbb{E}[\log |\mathcal{R}_S(\tilde{H}^-)|].$$

A similar mirror argument gives

$$P_{a:b \rightarrow c:d}(\lambda, e) \leq \mathbb{E}[\log |\mathcal{R}_A(H^-)|].$$

If $|\mathcal{R}_S(\tilde{h}^-)| \leq N_S$ uniformly, then

$$\mathbb{E}[\log |\mathcal{R}_S(\tilde{H}^-)|] \leq \log N_S.$$

Similarly, if $|\mathcal{R}_A(h^-)| \leq N_A$ uniformly, then

$$\mathbb{E}[\log |\mathcal{R}_A(H^-)|] \leq \log N_A.$$

Substituting the above bounds to get

$$E_{a:b \rightarrow c:d}(\lambda, e) \leq \log N_S, \quad P_{a:b \rightarrow c:d}(\lambda, e) \leq \log N_A.$$

Finally, by Corollaries 4.4 and 4.5,

$$L_U \leq E_{a:b \rightarrow c:d}(\lambda, e), \quad L_V \leq P_{a:b \rightarrow c:d}(\lambda, e).$$

Combining them with previous bounds,

$$L_U \leq \mathbb{E}[\log |\mathcal{R}_S(\tilde{H}^-)|], \quad L_V \leq \mathbb{E}[\log |\mathcal{R}_A(H^-)|].$$

Under the uniform support assumptions:

$$L_U \leq \log N_S, \quad L_V \leq \log N_A. \quad \square$$

Proof of Proposition 6.3. From Proposition 6.2,

$$E_{a:b \rightarrow c:d}(\lambda, e) \leq \mathbb{E}[\log |\mathcal{R}_S(\tilde{H}^-)|].$$

From Corollary 5.2, we also have

$$E_{a:b \rightarrow c:d}(\lambda, e) \leq (b - a + 1) \log |\mathcal{A}|.$$

Taking the minimum,

$$E_{a:b \rightarrow c:d}(\lambda, e) \leq \min \left\{ (b - a + 1) \log |\mathcal{A}|, \mathbb{E}[\log |\mathcal{R}_S(\tilde{H}^-)|] \right\}.$$

Similarly,

$$P_{a:b \rightarrow c:d}(\lambda, e) \leq \min \left\{ (b-a+1) \log |\mathcal{S}|, \mathbb{E}[\log |\mathcal{R}_A(H^-)|] \right\}.$$

Now compare the original bounds. For any history $\tilde{H}^- = \tilde{h}^-$, every reachable continuation $s_{c:d} \in \mathcal{R}_S(\tilde{h}^-)$ is some block of length $d-c+1$ whose entries in $\mathcal{S}$, i.e.

$$\mathcal{R}_S(\tilde{h}^-) \subseteq \mathcal{S}^{d-c+1},$$

so

$$|\mathcal{R}_S(\tilde{h}^-)| \leq |\mathcal{S}|^{d-c+1}.$$

Taking logs,

$$\log |\mathcal{R}_S(\tilde{h}^-)| \leq (d-c+1) \log |\mathcal{S}|.$$

Taking expectation,

$$\mathbb{E}[\log |\mathcal{R}_S(\tilde{H}^-)|] \leq (d-c+1) \log |\mathcal{S}|.$$

Similarly,

$$\mathbb{E}[\log |\mathcal{R}_A(H^-)|] \leq (d-c+1) \log |\mathcal{A}|.$$

Substituting these into the minimum bounds gives

$$E_{a:b \rightarrow c:d}(\lambda, e) \leq \min \left\{ (b-a+1) \log |\mathcal{A}|, (d-c+1) \log |\mathcal{S}| \right\},$$

$$P_{a:b \rightarrow c:d}(\lambda, e) \leq \min \left\{ (b-a+1) \log |\mathcal{S}|, (d-c+1) \log |\mathcal{A}| \right\}.$$

□

Proof of Example 6.4. Since the transitions are independent of actions and the policies are independent of states, we can factorize

$$\Pr(S_{1:n} = s_{1:n}, A_{1:n} = a_{1:n}) = \prod_{t=1}^n \frac{1}{|\mathcal{S}|} \cdot \prod_{t=1}^n \frac{1}{|\mathcal{A}|},$$

so $S_1, \dots, S_n, A_1, \dots, A_n$ are mutually independent, uniform random variables. Then for all $i \in \{c, \dots, d\}$,

$$H(S_i \mid A_{1:a-1}, S_{1:i-1}) = \log |\mathcal{S}|$$

and

$$H(S_i \mid A_{a:\min(b,i)}, A_{1:a-1}, S_{1:i-1}) = \log |\mathcal{S}|.$$

Therefore

$$I(A_{a:\min(b,i)}; S_i \mid A_{1:a-1}, S_{1:i-1}) = 0.$$

Every term in the expansion

$$E_{a:b \rightarrow c:d}(\lambda, e) = \sum_{i=c}^d I(A_{a:\min(b,i)}; S_i \mid A_{1:a-1}, S_{1:i-1})$$

is zero, so

$$E_{a:b \rightarrow c:d}(\lambda, e) = 0.$$

The same argument gives

$$P_{a:b \rightarrow c:d}(\lambda, e) = 0.$$

For the reachable-continuation sizes, fix any realized history $\tilde{H}^- = \tilde{h}^-$. Every block $s_{c:d} \in \mathcal{S}^{d-c+1}$ satisfies

$$\Pr(S_{c:d} = s_{c:d} \mid \tilde{H}^- = \tilde{h}^-) = |\mathcal{S}|^{-(d-c+1)} > 0.$$

Thus

$$\mathcal{R}_S(\tilde{h}^-) = \mathcal{S}^{d-c+1},$$

and so

$$\log |\mathcal{R}_S(\tilde{h}^-)| = (d-c+1) \log |\mathcal{S}|.$$

The action-side derivations are similar.

□

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